AN EXCELLENT SHIELD



- **An Active Electronically Scanned Array (AESA) radar provides situational awareness to a fighter jet by tracking targets, guiding missiles**
- **The DRDO's Uttam radar uses Transmit/ Receive Modules for faster beam switching, improved tracking, resistance to jamming**
- **An Electronic Warfare (EW) suite protects aircraft by jamming enemy radar, deploying countermeasures to evade detection/ attack**
- **The Swayam Raksha Kavach EW suite comprises the Radar Warning Receiver and the Advanced Self-Protection Jammer pod, providing cutting edge tech to the LCA Mk1A**

to integrating indigenous radar and EW systems for the last 43 aircraft, or from the 41st LCA Mk1A. However, in a volte face, HAL in March 2025 decided to put in radar and EW systems from ELTA, the supplier for the first 40 aircraft, for those final 43 aircraft as well. Though HAL was to start delivering LCA Mk1As from February 2024, not a single aircraft has materialised.

In a fighter jet, radars provide situational awareness by tracking targets and guiding missiles while an EW suite protects it from enemy radar by jamming them or deploying countermeasures. The AESA radar project, called the Uttam (or 'excellent') radar, was initiated in 2008 and sanctioned in 2012. DRDO completed the Transfer of Technology to HAL in July 2023, with Bharat Electronics Limited (BEL) as a subsystem supplier. The radar was cleared for export that year.

The SRK EW suite—designed to work with the Uttam radar—is being made by DRDO's Combat Aircraft Systems Development & Integration Centre, Bengaluru, and Defence Electronics Research Laboratory, Hyderabad. The EW suite comprises two components: the Radar Warning Receiver (RWR) and the Advanced Self-Protection Jammer (ASPJ) pod.

WHY FOREIGN CHOICE

An HAL official attributed the decision to the delayed certification of the DRDO's radar and EW suite by the Centre for Military Airworthiness and Certification (CEMILAC). "We have the imported radar as the DRDO radar and EW suite are not certified for production by CEMILAC completely," a key HAL official tells INDIA TODAY. HAL had issued a tender in November 2024 for 43 Uttam radars for the LCA Mk1A, making its U-turn all the more unexpected. A DRDO official acknowledges delays in conducting trials of the EW suite due to the non-availability of suitable test aircraft but claims that CEMILAC had certified the radar for production last year. The official adds that the DRDO had requested HAL for more time and that a final decision was yet to be taken.

DRDO officials point out a viable alternative—integrating the Uttam radar with imported EW systems until the SRK suite receives complete certification. In fact, making HAL's decision even more perplexing, in a recent internal communication to HAL and DRDO in April, CEMILAC stated that the Uttam radar had completed four phases of flight trials. Additionally, the radar hardware had undergone qualification tests according to the LCA Mk1A's requirements. To facilitate the induction of the radar, CEMILAC recommended commencing production of the radar subsystems, with HAL's avionics division in Hyderabad serving as the lead system integrator.

A senior DRDO official suggests that HAL's hesitation may stem from a lack of confidence in the indigenous system. "It seems it is under pressure to deliver the jets," he tells INDIA TODAY. "If the air force has expressed confidence, the HAL should also be a little patient to adopt an Indian made system. Moreover, by the time HAL reaches at 41 aircraft, the Indian EW suite would be certified," he says. HAL's decision to opt for imported technology, he adds, undermines the efforts towards indigenisation. Ravi Gupta, a former DRDO scientist, states that the DRDO's Electronics and Radar Development Establishment (LRDE) invested years in developing these technologies. "Having complete control over the supply chain and source code of indigenous radar systems is essential for national security, particularly in the event of war, when imported systems may be compromised," he says.

The LRDE co-developed the Uttam AESA radar system with other DRDO labs and partners like BEL. Unlike mechanically scanned radars, Uttam employs electronically controlled antennas, known as Transmit/ Receive Modules (TRMs), which enable faster beam switching and improved target tracking and resistance to jamming.

The Uttam radar has also demonstrated an ability to track up to 50 airborne targets at ranges beyond 100 km and engage four targets simultaneously. The IAF has taken a keen interest in the Uttam radar, and top IAF leadership has praised its performance.

However, a senior DRDO official says that flight trials of the radar and development of the EW suite faced numerous hurdles. Developing cutting-edge systems from scratch with industry partners is a huge challenge. Secondly, DRDO's access to only older Tejas LCA aircraft hindered progress. In contrast, foreign manufacturers were provided with two LCA Mk1A prototypes.

The following weeks will reveal whether HAL's decision will stand, or if the DRDO's painstaking efforts will be honoured. One thing is clear—the IAF, MoD officials and most Indian scientists are standing by the Uttam radar and the SRK EW suite. ■